

PHYSICS – BOARD QUESTION PAPER : FEBRUARY 2024**Time: 3 Hours****Max. Marks: 70****General Instructions:**

1. Section A: Q1 contains Ten multiple choice questions. Q2 contains Eight very short questions.
2. Section B: Q3 to Q14 contain Twelve questions. Attempt any Eight.
3. Section C: Q15 to Q26 contain Twelve questions. Attempt any Eight.
4. Section D: Q27 to Q31 contain Five questions. Attempt any Three.
5. Use of log table is allowed. Calculator is not allowed.
6. Figures to the right indicate full marks.
7. Only the first attempt for MCQs will be considered.

Physical Constants:Mass of electron = 9.1×10^{-31} kg $\epsilon_0 = 8.85 \times 10^{-12}$ C²/Nm² $\pi = 3.142$ Charge on electron = 1.6×10^{-19} C $\mu_0 = 4\pi \times 10^{-7}$ Wb/AmPlanck's constant = 6.63×10^{-34} JsSpeed of light = 3×10^8 m/s $g = 9.8$ m/s²Rydberg's constant = 1.097×10^7 m⁻¹Stefan's constant = 5.67×10^{-8} J m⁻² s⁻¹ K⁻⁴**SECTION – A****Q.1 Select and write the correct answers: (10M)**

i. The moment of inertia of a disc about its central axis is:

(A) $\frac{1}{2}MR^2$ (B) $\frac{1}{4}MR^2$ (C) MR^2 (D) $\frac{3}{2}MR^2$

ii. Dimensional formula of surface tension:

(A) $[L^{-1} M T^{-2}]$ (B) $[L^2 M T^{-2}]$ (C) $[L M T^{-1}]$ (D) $[M T^{-2}]$

iii. Phase difference between node and adjacent antinode:

(A) $\pi/4$ (B) $\pi/2$ (C) $3\pi/4$ (D) π 

iv. Work done in bringing unit charge from infinity to a point:

(A) Electric flux (B) Magnetic potential (C) Electric potential (D) Gravitational potential

v. To convert galvanometer into ammeter, connect:

(A) Small resistance parallel (B) Large resistance series

(C) Small resistance series (D) Large resistance parallel

vi. If frequency doubles, kinetic energy becomes:

(A) Same (B) Twice (C) More than twice (D) Less than twice

vii. In cyclic process:

(A) $\Delta U = Q$ (B) $Q = 0$ (C) $W = 0$ (D) $W = Q$

viii. Self inductance 20H, current 50A $\rightarrow$ 10A in 0.1s. Induced emf:

(A) 800V (B) 6000V (C) 7000V (D) 8000V

ix. Velocity of bob at 6 cm, amplitude 10 cm:

(A) 8π cm/s (B) 6π cm/s (C) 4π cm/s (D) 2π cm/s

x. In biprism, 20th fringe = 1.2 cm. 30th fringe =

(A) 0.6 cm (B) 0.8 cm (C) 1.2 cm (D) 1.8 cm

Q.2 Answer the following: (8M)

i. Define centripetal force.

ii. Why is detergent mixed with water?

iii. Resistance of an ideal voltmeter.

iv. Write vector formula of torque on current-carrying coil.

v. What is binding energy of hydrogen atom?

vi. Define surroundings in thermodynamics.

vii. Stopping potential = 1.5V. Find maximum kinetic energy.

viii. Capacitors $5\mu\text{F}$ & $10\mu\text{F}$ in series. Find equivalent capacitance.

SECTION – B

Attempt any Eight questions. (16M)

Q3. Explain change in internal energy of gas by heating.

Q4. Explain construction of spherical wavefront using Huygens' principle.

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- Q5. Define magnetization. State SI unit and dimensions.
- Q6. Obtain differential equation of linear SHM.
- Q7. Galvanometer 30Ω , $20\ \mu\text{A}$. Find series resistance for 0–10 V range.
- Q8. Explain Biot–Savart law.
- Q9. What is LED? Draw symbol.
- Q10. Wingspan 60 m, $B = 6 \times 10^{-5}\ \text{T}$, speed = 500 m/s. Find induced emf.
- Q11. Derive expression for maximum speed on horizontal circular track.
- Q12. A 0.5 N force moves metal plate on glycerin layer. Find viscosity.
- Q13. Tuning forks 320 & 340 Hz, velocity 340 m/s. Find wavelength difference.
- Q14. Change in angular momentum from $n=3$ to $n=1$ orbit.

SECTION – C

Attempt any Eight questions. (24M)

- Q15. Coil 200 turns, radius 10 cm, current 0.5A. Find magnetic field at center.
- Q16. Define photoelectric effect and explain experiment.
- Q17. Define α_{DC} and β_{DC} . Obtain relation.
- Q18. Define surface energy and derive relation with surface tension.
- Q19. What is isothermal process? Derive work done.
- Q20. Derive equation of stationary wave. Show node spacing $\lambda/2$.
- Q21. Derive impedance of LCR circuit. Draw phasor diagram.
- Q22. Find wavelengths of first two lines of Balmer series.
- Q23. Toroid filled with lithium, susceptibility 2.1×10^{-5} . Find % increase in field.
- Q24. Track radius 200 m. Max speed 25 m/s. Find banking angle.
- Q25. Prove Mayer's relation $C_p - C_v = R/J$.
- Q26. $e = 8 \sin 628.4t$. Find peak value, frequency, and value at $t = 10\ \text{ms}$.

SECTION – D

Attempt any Three questions. (12M)

- Q27. What is transformer? Explain construction, working and derive equation.
- Q28. Using double slit geometry, derive expression for fringe width.
- Q29. Distinguish between ammeter and voltmeter. A particle at $x = A/3$. Find fraction of KE.
- Q30. Draw Ferry's black body. Compare rms speeds of H_2 at 227°C and O_2 at 127°C .
- Q31. Derive energy stored in capacitor. A sphere radius 15 cm carries $2\ \mu\text{C}$. Find electric field at 20 cm.

